

Supplementary Table 1: The Conjoint Triad results on 25 datasets using our proposed model.

Dataset	Accuracy	Precision	Recall	F1-Score	Specificity	MCC	PR	ROC
Data_1	86.4	86.08	86.8	86.44	86	72.81	92.32	93.28
Data_2	85.92	84.03	88.72	86.31	83.18	72.01	91.68	92.79
Data_3	85.03	85.17	84.79	84.98	85.27	70.07	90.69	92.09
Data_4	85.41	84.68	86.4	85.53	84.41	70.83	91.03	92.27
Data_5	86.39	84.93	88.44	86.65	84.35	72.85	91.28	92.48
Data_6	86.37	86.34	86.36	86.35	86.37	72.74	91.75	93.02
Data_7	86.53	86.01	87.22	86.61	85.84	73.08	91.46	92.89
Data_8	86.43	86.29	86.58	86.44	86.29	72.87	92.21	93.16
Data_9	86.08	85.94	86.23	86.08	85.93	72.16	90.83	92.62
Data_10	86.66	86.07	87.44	86.75	85.89	73.34	92.3	93.04
Data_11	85.98	84.35	88.3	86.28	83.66	72.05	91.55	92.57
Data_12	85.89	86.34	85.23	85.78	86.55	71.79	91.51	92.44
Data_13	85.38	84.48	86.65	85.55	84.13	70.8	90.94	92.34
Data_14	86.61	86.49	86.74	86.61	86.48	73.22	91.8	92.97
Data_15	86.43	85.96	87.04	86.5	85.82	72.88	92.02	93.01
Data_16	85.73	84.72	87.13	85.91	84.32	71.49	91.39	92.53
Data_17	86.02	85.83	86.25	86.04	85.8	72.06	91.17	92.43
Data_18	85.99	86.53	85.21	85.87	86.77	72	91.23	92.48
Data_19	86.34	86.54	86.03	86.29	86.66	72.7	92.07	92.99
Data_20	86.42	86.52	86.25	86.38	86.59	72.85	90.92	92.52
Data_21	86.89	86.1	87.93	87.01	85.84	73.79	91.83	93.01
Data_22	87.14	86.8	87.55	87.18	86.73	74.29	92.07	93.33
Data_23	86.6	85.47	88.15	86.79	88.05	73.24	92.26	93.06
Data_24	86.28	87.14	85.08	86.1	87.48	72.58	91.43	92.71
Data_25	86.11	87.15	94.3	90.58	66.15	64.97	93.67	89.78

Supplementary Table 2: The amino acid sequence-based features result on 25 datasets using our proposed model.

Dataset	Accuracy	Precision	Recall	F1-Score	Specificity	MCC	PR	ROC
Data_1	50.51	50.42	52.15	51.27	48.86	40.38	63.23	50.5
Data_2	50.4	50.33	49.08	49.7	51.7	39.53	62.42	50.39
Data_3	50.21	50.14	50.43	50.28	49.98	41.99	62.66	50.2
Data_4	49.66	49.59	50.05	49.82	49.26	48.31	62.29	49.65
Data_5	49.61	49.53	47.77	48.63	51.44	40.25	61.69	49.61
Data_6	49.46	49.38	48.77	49.07	50.14	42.48	61.86	49.45
Data_7	50.04	49.97	50.25	50.11	49.83	43.29	62.53	50.04
Data_8	51.16	51.26	44.46	47.62	57.83	43.82	61.72	51.14
Data_9	51.07	50.94	54.18	52.51	47.96	41.01	64	51.07
Data_10	49.49	49.39	46.69	48	52.28	44.45	61.35	49.48
Data_11	49.27	49.22	50.29	49.75	48.24	40.9	62.16	49.27
Data_12	50.53	50.46	50.49	50.48	50.56	41.78	62.83	50.52
Data_13	50.52	50.43	52.77	51.57	48.26	40.65	63.39	50.52
Data_14	49.59	49.5	47.64	48.55	51.53	41.46	61.64	49.58
Data_15	49.37	49.31	50.2	49.75	48.53	42.36	62.19	49.37
Data_16	49.81	49.74	50.43	50.08	49.19	44.18	62.46	49.81
Data_17	50.36	50.28	51.84	51.05	48.88	43.57	63.09	50.36
Data_18	50.21	50.15	47.46	48.77	52.94	41.27	61.92	50.2
Data_19	50.74	50.63	53.54	52.05	47.93	48.84	63.69	50.74
Data_20	51.53	51.46	51.62	51.54	51.44	45.24	63.62	51.53
Data_21	50.58	50.51	51.31	50.9	49.85	44.38	63.06	50.58
Data_22	50.61	50.54	50.58	50.56	50.62	46.76	62.9	50.6
Data_23	50.01	49.94	48.75	49.34	51.26	39.82	62.14	50
Data_24	49.94	49.87	49.45	49.66	50.42	40.14	62.28	49.94
Data_25	50.08	50.21	50.67	49.92	50.2	44.69	61.34	49.98

Supplementary Table 3: The GO terms result on 25 datasets using our proposed model.

Dataset	Accuracy	Precision	Recall	F1-Score	Specificity	MCC	PR	ROC
Data_1	67.53	68.39	64.43	66.35	70.6	35.1	72.58	73.59
Data_2	81.33	80.37	82.59	81.47	80.09	62.7	86.62	87.76
Data_3	80.92	79.9	82.29	81.08	79.57	61.87	85.22	87.54
Data_4	80.49	79.29	82.44	80.83	78.54	61.03	85.07	87.11
Data_5	81.16	81.87	79.73	80.78	82.56	62.33	86.97	87.87
Data_6	81.78	82.87	79.83	81.32	83.71	63.61	86.18	87.98
Data_7	80.96	80.06	82.11	81.07	79.82	61.94	86.12	87.56
Data_8	81.45	80.45	82.77	81.59	80.14	62.93	86.79	88.44
Data_9	80.65	79.12	82.94	80.99	78.39	61.39	85.51	87.16
Data_10	80.7	81.53	79.07	80.28	82.31	61.43	86.19	87.39
Data_11	80.29	84.31	74.12	78.89	86.38	60.99	86.94	87.49
Data_12	80.59	79.77	81.63	80.69	79.57	61.21	85.41	87.31
Data_13	80.8	82.96	77.23	79.99	84.34	61.74	85.6	87.34
Data_14	81.36	80.96	81.68	81.32	81.04	62.72	86.89	87.86
Data_15	81.26	82.3	79.33	80.79	83.16	62.55	85.33	87.49
Data_16	80.48	79.76	81.35	80.55	79.62	60.98	84.49	86.56
Data_17	82.27	81.77	82.77	82.27	81.79	64.56	87.4	88.68
Data_18	80.98	83.29	77.2	80.13	84.71	62.11	87.19	88.15
Data_19	81.65	81.66	81.33	81.49	81.96	63.3	87.07	88
Data_20	83.03	83.62	81.88	82.74	84.16	66.07	87.34	89.41
Data_21	80.68	78.17	84.79	81.34	76.62	61.6	85.87	87.84
Data_22	81.58	84.82	76.65	80.53	86.46	63.44	87.1	88.32
Data_23	81.71	83.57	78.64	81.03	84.74	63.52	87.03	88.24
Data_24	82.28	83.02	83.47	83.43	80.51	64.4	87.44	88.51